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Unit 3 : Session 10 : SLO 2 — Tutorial: Non-Restoring Division

■ Problem 2: Non-Restoring Division of 0101 (5) ÷ 0010 (2)

- Init: $A = 0000$, $Q = 0101$, $M = 0010$
- Step 1: shift (A,Q) → $A = 0000$, $Q = 1010$; $A \geq 0 \rightarrow A = A - M = 1110 (< 0)$, $Q_0 = 0 \rightarrow A = 1110$, $Q = 1010$
- Step 2: shift → $A = 1101$, $Q = 0100$; $A < 0 \rightarrow A = A + M = 1101 + 0010 = 1111 (< 0)$, $Q_0 = 0 \rightarrow A = 1111$, $Q = 0100$
- Step 3: shift → $A = 1110$, $Q = 1000$; $A < 0 \rightarrow A = A + M = 1110 + 0010 = 0000 (\geq 0)$, $Q_0 = 1 \rightarrow A = 0000$, $Q = 1001$
- Step 4: shift → $A = 0001$, $Q = 0010$; $A \geq 0 \rightarrow A = A - M = 0001 - 0010 = 1111 (< 0)$, $Q_0 = 0 \rightarrow A = 1111$, $Q = 0010$
- Final correction: $A < 0 \rightarrow A = A + M = 1111 + 0010 = 0001$
- Final Quotient = $Q = 0010$ (binary) = 2 (decimal)
- Final Remainder = $A = 0001$ (binary) = 1 (decimal)
- Verification: $5 \div 2 = 2$ remainder 1 ✓